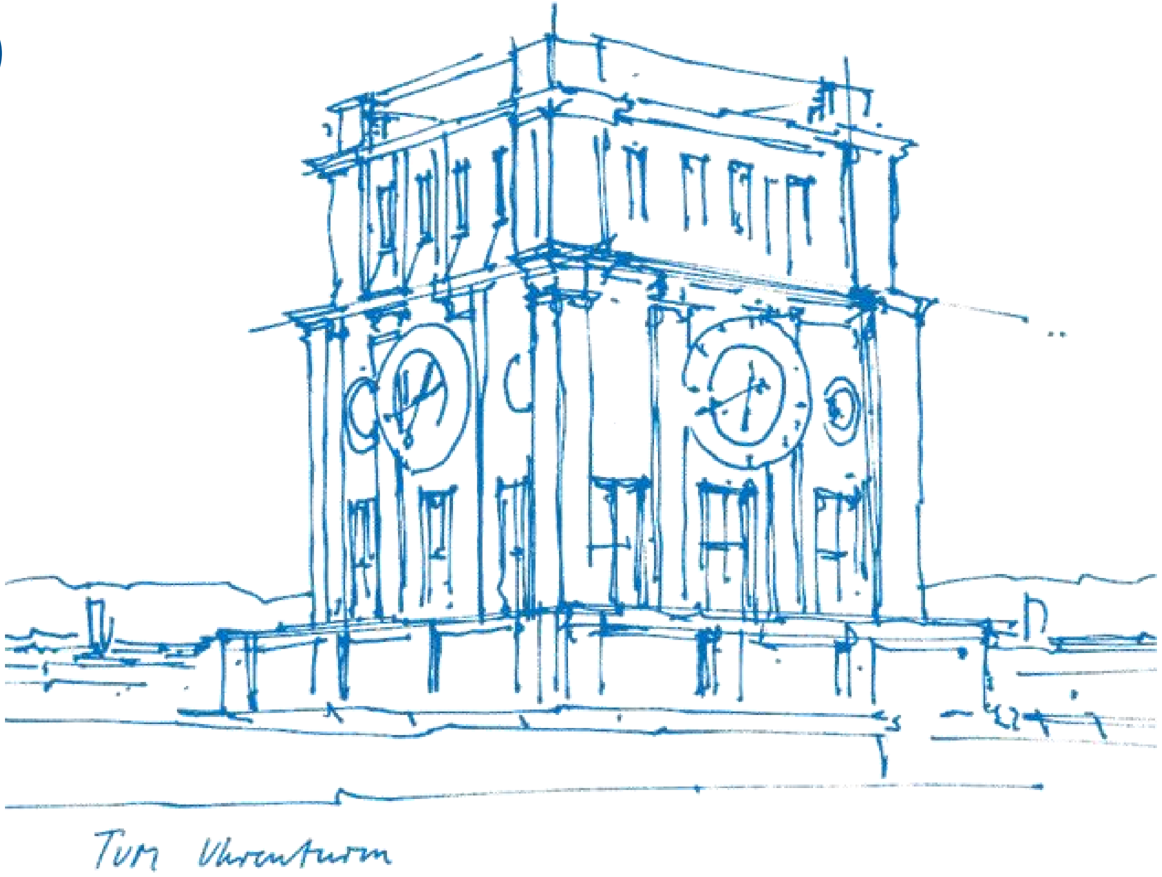
Discrete Probability Theory (IN0018)
School of Computation, Information, and Technology
Technical University of Munich

Tutorial 10

Transformation of Densities, Maximum Likelihood Estimation (mle)



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About Tutorial

- **Tutorial:** In-Person, 2h by two time periods per week - 12:00-14:00 Tue, 03.11.018 (Group B33)
  - 10:00-12:00 Fri, 01.06.011 (Group E24)

- The slides are not guaranteed to be accurate! (Written by myself)

- **Materials of this tutorial:** Please scan the QR-Code ☞

➢ PART I. Recap of the Lecture

➢ PART II. Exercises from the Exercise Sheet

➢ PART III. Additional Exercise (with solution)

https://drive.google.com/drive/folders1rBeL0NKHuYzrULK3Fr4CGr8svRhyTNOO?usp=sharing

I. Recap

- **(Review)**
  - ① - Characteristic function
  - ② - Central Limit Theorem
  - ③ - Convergence (in distribution)

- **Transformation of densities / changes of variables**
  - ④ - Definition

- **Maximum Likelihood Estimation (mle)**
  - ⑤ -

For dataset $x_1, x_2 \cdots x_n$, assume they come from a distribution with parameter θ.

MLE finds the value $\hat{\theta}$ makes the observed data most likely, $L(\theta | x)$ achieves maximum value.

↳ $\hat{\theta}_{MLE} = \arg\max L(\theta | x)$.

In that $L(\theta | x) = p(x_1, x_2, \cdots x_n | \theta)$ "Likelihood of θ given x".

e.g. Flipping coin (not average) 10 times, 7 heads, 3 tails.

To estimate: "Prob. of head". $L(\theta) = \theta^7 (1-\theta)^3$

Find $\hat{\theta}$ such that $L(\theta)$ gets the maximum!

$\Rightarrow \hat{\theta} = 0.7$

⑥ - Unbiased Estimator

I. Recap

Normal distribution table

e.g. $\Phi(1.56)=0.941$

$\Phi(2.82)=0.998$

$\Phi(x)$	0,00	0,01	0,02	0,03	0,04	0,05	0,06	0,07	0,08	0,09
0,0	0,500	0,504	0,508	0,512	0,516	0,520	0,524	0,528	0,532	0,536
0,1	0,540	0,544	0,548	0,552	0,556	0,560	0,564	0,567	0,571	0,575
0,2	0,579	0,583	0,587	0,591	0,595	0,599	0,603	0,606	0,610	0,614
0,3	0,618	0,622	0,626	0,629	0,633	0,637	0,641	0,644	0,648	0,652
0,4	0,655	0,659	0,663	0,666	0,670	0,674	0,677	0,681	0,684	0,688
0,5	0,691	0,695	0,698	0,702	0,705	0,709	0,712	0,716	0,719	0,722
0,6	0,726	0,729	0,732	0,736	0,739	0,742	0,745	0,749	0,752	0,755
0,7	0,758	0,761	0,764	0,767	0,770	0,773	0,776	0,779	0,782	0,785
0,8	0,788	0,791	0,794	0,797	0,800	0,802	0,805	0,808	0,811	0,813
0,9	0,816	0,819	0,821	0,824	0,826	0,829	0,831	0,834	0,836	0,839
1,0	0,841	0,844	0,846	0,848	0,851	0,853	0,855	0,858	0,860	0,862
1,1	0,864	0,867	0,869	0,871	0,873	0,875	0,877	0,879	0,881	0,883
1,2	0,885	0,887	0,889	0,891	0,893	0,894	0,896	0,898	0,900	0,901
1,3	0,903	0,905	0,907	0,908	0,910	0,911	0,913	0,915	0,916	0,918
1,4	0,919	0,921	0,922	0,924	0,925	0,926	0,928	0,929	0,931	0,932
1,5	0,933	0,934	0,936	0,937	0,938	0,939	0,941	0,942	0,943	0,944
1,6	0,945	0,946	0,947	0,948	0,949	0,951	0,952	0,953	0,954	0,954
1,7	0,955	0,956	0,957	0,958	0,959	0,960	0,961	0,962	0,962	0,963
1,8	0,964	0,965	0,966	0,966	0,967	0,968	0,969	0,969	0,970	0,971
1,9	0,971	0,972	0,973	0,973	0,974	0,974	0,975	0,976	0,976	0,977
2,0	0,977	0,978	0,978	0,979	0,979	0,980	0,980	0,981	0,981	0,982
2,1	0,982	0,983	0,983	0,983	0,984	0,984	0,985	0,985	0,985	0,986
2,2	0,986	0,986	0,987	0,987	0,987	0,988	0,988	0,988	0,989	0,989
2,3	0,989	0,990	0,990	0,990	0,990	0,991	0,991	0,991	0,991	0,992
2,4	0,992	0,992	0,992	0,992	0,993	0,993	0,993	0,993	0,993	0,994
2,5	0,994	0,994	0,994	0,994	0,994	0,995	0,995	0,995	0,995	0,995
2,6	0,995	0,995	0,996	0,996	0,996	0,996	0,996	0,996	0,996	0,996
2,7	0,997	0,997	0,997	0,997	0,997	0,997	0,997	0,997	0,997	0,997
2,8	0,997	0,998	0,998	0,998	0,998	0,998	0,998	0,998	0,998	0,998
2,9	0,998	0,998	0,998	0,998	0,998	0,998	0,998	0,999	0,999	0,999

Exercise 1 (sum of waiting times). The time (in minutes) a customer i has to wait for their coffee at a specialty coffee shop is modeled by an exponential random variable $X_i \sim \mathrm{Exp}(\lambda)$ with $\lambda = 0.5$ (so the average wait is $1/\lambda = 2$ minutes). Let $S_{36} = \sum_{i=1}^{36} X_i$ be the total time required to serve 36 *independent* customers.

(i) Find the characteristic function of a single waiting time X_i.

(ii) Using the result from (i), what is the exact distribution of the total waiting time S_{36}? State its name and parameters.
[Hint: Look at Example 2.50 of the lecture.]

(iii) Use the clt to approximate the probability that the total waiting time for 36 customers exceeds 80 minutes.
[Hint: You will need the mean and variance of an $\mathrm{Exp}(0.5)$ distribution. For the clt, you may use that for a standard normal RVRV Z, we have $P(Z > 2/3) \approx 0.2525$.]

Exercise 2 (quality control in manufacturing). A factory produces resistors. The resistance of each resistor is intended to be 100 ohms. Due to manufacturing variability, the actual resistance values are iid random variables with mean $\mu = 100$ ohms and standard deviation $\sigma = 2$ ohms. A batch consists of $n = 64$ resistors. A batch is rejected if the average resistance of the 64 resistors is outside the range $[99.5, 100.5]$ ohms.

(i) Let $\bar{X}_{64}$ be the average resistance of a batch of 64 resistors. What are the mean and variance of $\bar{X}_{64}$?

(ii) Use Chebyshev's inequality to find an upper bound on the probability that a batch is rejected.

(iii) Use the clt to approximate the probability that a batch is rejected.
[Hint: You may use that for a standard normal RVRV Z, we have $P(Z > 2) \approx 0.0228$.]

II. Exercise Sheet

Exercise 3 (transformation and convergence in distribution). Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of RVRVs with $X_n \sim \mathrm{Unif}(0, 1/n)$ for each $n \in \mathbb{N}_{>0}$. Let $Y_n = nX_n$.

(i) For a fixed n, find the pdf $p_{Y_n}(y)$ of Y_n.
[Hint: Use the change of variables formula.]

(ii) Find the cdf $F_{Y_n}(y)$ of Y_n.

(iii) Determine the limiting cdf $F_Y(y) = \lim_{n\to\infty} F_{Y_n}(y)$.

(iv) To what RVRV Y does Y_n converge in distribution? Specify its distribution.

II. Exercise Sheet

Exercise 4 (mle for uniform distribution). Let $X_1, \dots, X_n$ be iid RVRVs from a uniform distribution $\mathrm{Unif}(0, \theta)$ with unknown parameter $\theta > 0$. The pdf of X_i is $p(x_i \mid \theta) = \frac{1}{\theta}\chi_{[0,\theta]}(x_i)$.

(i) Write down the likelihood function $L(\theta \mid x)$ for an observed sample $x = (x_1, \dots, x_n)$. Be careful with the indicator functions.

(ii) Find the maximum likelihood estimator (mle) $\hat{\theta}_{\mathrm{mle}}$ for θ.
[Hint: The likelihood function may not be differentiable with respect to θ, but the maximum value can be found by reasoning about monotonicity and where it is non-zero.]

(iii) Calculate the expected value $\mathbb{E}[\hat{\theta}_{\mathrm{mle}}]$. Is $\hat{\theta}_{\mathrm{mle}}$ an unbiased estimator for θ?
[Hint: First find the cdf of $\hat{\theta}_{\mathrm{mle}}$, then its pdf, and then compute the expectation.]

III. Additional Exersice

Exercise 5. The student Arnold wants to estimate the number n of fish in the lake behind the Faculty of Mechanical Engineering building. He devises the following method:

1. First, he catches a random sample of m fish, marks them appropriately, and releases them back into the lake.
2. After some time, he catches another random sample of k fish.
3. Let X be the number of marked fish in the second sample.

Objective: Determine a maximum likelihood estimator (MLE) for n based on X.
Hint: For optimizing the likelihood function, consider the ratio of two consecutive values of n, i.e.,

$$\frac{L(x;n)}{L(x;n-1)}.$$

III. Additional Exersice

Intuitiv ist X Hypergeometrische-verteilt. $P(X=x) = \frac{\binom{m}{x}\binom{n-m}{k-x}}{\binom{n}{k}}$.

Aus Hinweis:

$$\frac{L(x;n)}{L(x;n-1)} = \frac{\binom{m}{x}\binom{n-m}{k-x}}{\binom{n}{k}} \cdot \frac{\binom{n-1}{k}}{\binom{m}{x}\binom{n-1-m}{k-x}} = \frac{\binom{n-m}{k-x}\binom{n-1}{k}}{\binom{n}{k}\binom{n-1-m}{k-x}}$$

$$= \frac{\cancel{(n-m)!}}{\cancel{(k-x)!}\,\cancel{(n-m-k+x)!}} \cdot \frac{\cancel{(n-1)!}}{\cancel{k!}\,\cancel{(n-1-k)!}} \cdot \frac{\cancel{k!}\,\cancel{(n-k)!}}{\cancel{n!}} \cdot \frac{\cancel{(k-x)!}\,\cancel{(n-1-m-k+x)!}}{\cancel{(n-1-m)!}}$$

$$= \frac{(n-m)\,(n-k)}{n\,(n-m-k+x)} = \frac{n^2-nk-mn+mk}{n^2-nm-nk+nx} \le 1,\quad n^2-nk-mn+mk \le n^2-nm-nk+nx,\quad \hat{n} = \frac{m\cdot k}{x}.$$